

Name:

Sticker

Matric Number:

Tutorial Group:

Constants and Formula List

Gravitational Acceleration	g	9.8 m/s ²
Mass of Earth	M _e	5.97×10 ²⁴ kg
Gravitational constant	G	6.67×10 ⁻¹¹ N·m ² /kg ²
Avogadro's number	N _A	6.02×10 ²³ molecules/mol
Universal gas constant	R	8.314 J/mol K
Absolute zero	0 K	- 273.15°C
Pressure conversion	1 bar	10 ⁵ N/m ² (or Pa)
	1 atm	1.013 × 10 ⁵ Pa

Basics and Thermal Physics

$$\Delta L = \alpha L_i \Delta T, \quad \Delta V = \beta V_i \Delta T, \quad PV = nRT, \quad Q = mc\Delta T, \quad Q = ml$$

$$T = \frac{P(T) - P(0)}{P(100) - P(0)} \times 100, \quad T = 273.16 \text{ K } \lim_{P_{tr} \rightarrow 0} \frac{P}{P_{tr}}$$

Linear Motion

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_f - \vec{r}_i}{t_f - t_i}, \quad \vec{v}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}, \quad v = \frac{dx}{dt}, \quad \text{Gravitation Potential Energy} = mgh; \text{ Kinetic Energy} = \frac{1}{2}mv^2$$

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}, \quad \vec{a}_{int} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}, \quad a = \frac{dv}{dt}$$

$$v_f = v_i + at, \quad x_f = x_i + \frac{1}{2}(v_i + v_f)t, \quad v_f^2 = v_i^2 + 2a(x_f - x_i), \quad x_f = x_i + v_it + \frac{1}{2}at^2$$

$$y = (\tan \theta_i)x - \left(\frac{g}{2v_i^2 \cos^2 \theta_i} \right) x^2 \quad (\text{taking downwards as } -ve, \quad \theta_i \text{ is measured above } +ve \text{ } x \text{ axis}),$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}, \quad R = \frac{v_i^2 \sin 2\theta_i}{g}$$

$$\Sigma \vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}, \quad \vec{p} = m\vec{v}, \quad f_k = \mu_k N, \quad f_s \leq \mu_s N, \quad |F_{sp}| = kx, \quad v = \frac{mg}{b} \left(1 - e^{-\frac{b}{m}t} \right)$$

Rotational Motion

$$a_c = \frac{v^2}{r} = r\omega^2, \quad a_t = \frac{d|\vec{v}|}{dt}, \quad s = r\theta, \quad \omega = \frac{d\theta}{dt} = \frac{v}{r}, \quad \alpha = \frac{d\omega}{dt} = \frac{a}{r}; \quad \omega_f = \omega_i + \alpha t, \\ \theta_f = \theta_i + \frac{1}{2}(\omega_i + \omega_f)t, \quad \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i), \quad \theta_f = \theta_i + \omega_it + \frac{1}{2}\alpha t^2$$

Mathematics

$$\tan A = \frac{\sin A}{\cos A}; \quad \sin^2 A + \cos^2 A = 1; \quad \sec^2 A = \frac{1}{\cos^2 A} = \tan^2 A + 1; \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ where } ax^2 + bx + c = 0; \\ \frac{d}{dx}(ax^n) = anx^{n-1}; \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1); \quad \int \frac{dx}{x} = \ln x$$

Instructions:

- a) Answer all questions.
- b) Working for all questions must be shown clearly in the space provided.

Short Questions (31 points)

1. [3 points] There are two units to represent fuel consumption by a car: volume of fuel needed to travel 100 km; or distance travelled for a unit volume of fuel. A car's fuel consumption is reported as 7.50 L / 100 km. Convert this to miles per US gallon. You are given that 1 mile=1.609 km and 1 US gallon = 3.785 L. Leave your answer to 1 d.p.

ANS:

2. [3 points] The pressure inside an airplane is maintained at 0.81 atm. Determine the net outward force (in N) exerted on a 1.10 m x 1.83 m cabin door if the outside pressure (at 10 km height) is 0.27 atm. Leave your answer to the nearest 100N.

ANS:

3. [2 points] A boat is floating in a small pond. The boat then sinks so that it is completely submerged. What happens to the level of the pond?

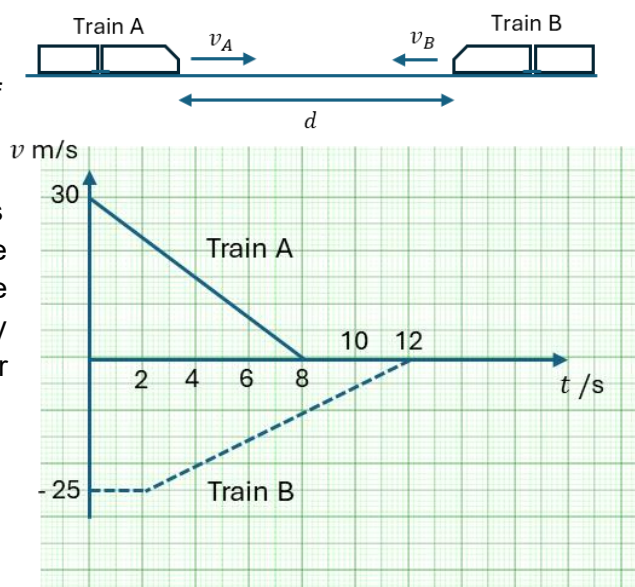
- a. It increases.
- b. It decreases.
- c. It stays the same.
- d. In some cases it would decrease, in others it would stay the same.
- e. In some cases it would increase, in others it would stay the same.

ANS:

4. [3 points] A bag of potato chips contains 2.00 L of air when it is sealed at sea level at a pressure of 1.00 atm and a temperature of 20.0°C. What will be the volume of the air in the bag if you take it with you, still sealed, to the mountains where the temperature is 7.00°C and atmospheric pressure is 70.0 kPa? (Assume the air in the bag is in thermal equilibrium with the outside air.)

ANS:

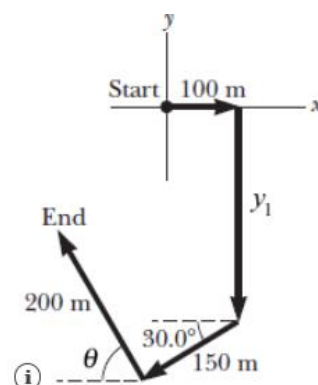
5. [4 points] As two trains move along the same track when they are d metres apart, the driver of train A realized that he has to apply brakes immediately so as not to collide with train B. The driver of train B apply the brakes 2 seconds later than the driver of train A. The velocity time graph of the two trains is given. The trains are only 20 m apart when both trains fully stop. Determine the distance d . Leave your answer to the nearest m.



ANS:

 m

6. [5 points] A person going for a walk follows the path shown in the figure, where $y_1 = 266$ m and $\theta = 64.0^\circ$. The total trip consists of four straight-line paths. At the end of the walk, what is the person's resultant displacement measured from the starting point? Leave your answers to 1 d.p.



ANS: Magnitude:

 m

Direction:

 °

Matric Number:

7. [5 points] The position **A** of an object with respect to the origin **O** of a two-dimensional system at time t is given by

$$\vec{r}_{AO} = (0.15 + 0.12t - 0.03t^2)\hat{i} + (-0.96t + 0.08t^3)\hat{j}$$

where $\vec{r}$ is in meters (m) and t in seconds (s). The object has a mass of 1.2 kg.

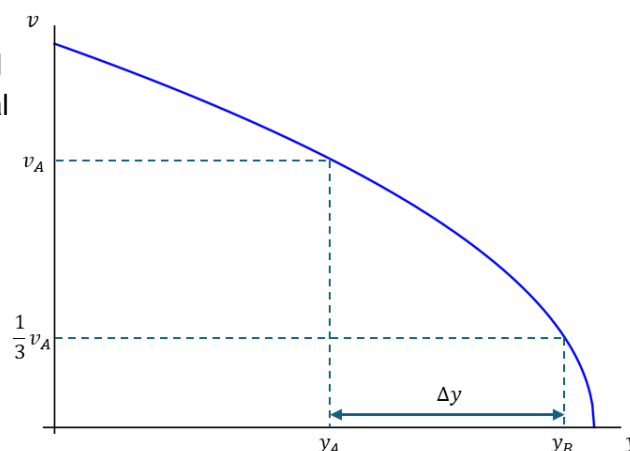
- Determine the time t_1 that the object comes to rest momentarily (i.e. velocities in both directions are zero.).
- Calculate the net force (in vector form) acting on the object at time $t = 2.5$ s. Leave your answers to 2 d.p.

ANS: $t_1 =$ s $F =$ N

8. [3 points] An object of mass 0.30 kg at rest is released from a height of 8.0 m above the ground. Calculate the change in gravitational potential energy when the ball falls to just 3.0 m above the ground. Calculate the velocity of the ball when it is at 3.0 m above the ground.

ANS: J m/s

9. [3 points] A ball is thrown vertically upwards. The figure on the right shows how the vertical speed v of the ball varies with the vertical distance y from the point it leaves the hand. Given that the speed at y_B is $\frac{1}{3}v_A$ and that $\Delta y = 0.60$ m. Determine v_A .



ANS: m/s

Structured Questions (29 points)

Q1. [8 points]

- a) If a 600 MW data center requires 50 days of computation to train an AI model, determine the power of the data center (in MW) needed to complete the training of the same AI model in 30 days. Leave your answer to the nearest 10 MW.

ANS:

MW

- b) An AI training computer server uses a closed-loop liquid cooling system with a manifold feeding identical cold plates mounted on each chip. that heat isn't spread evenly across the machine's entire mass. The vast majority is generated by the GPUs and CPUs in a very small area. Assuming that this small area is generating heat at $8\,000\text{ J / s}$ resulting it to rise by an estimated 15.0°C every second and it is cooled by a liquid cooling system. You can treat the liquid as water. Assuming that there are no other types of heat losses and the cooling efficiency is 100%. Given that for water, the specific heat capacity = $4186\text{ J / kg }^\circ\text{C}$; latent heat of fusion = 333 kJ / kg ; latent heat of vaporization = 2260 kJ / kg . As a comparison, the specific heat capacity of air is about $1000\text{ J / kg }^\circ\text{C}$.)
- i. Determine the mass flow rate water (in grams per second) needed to reduce the temperature by 15.0°C every second, thus maintaining the temperature of the hardware.

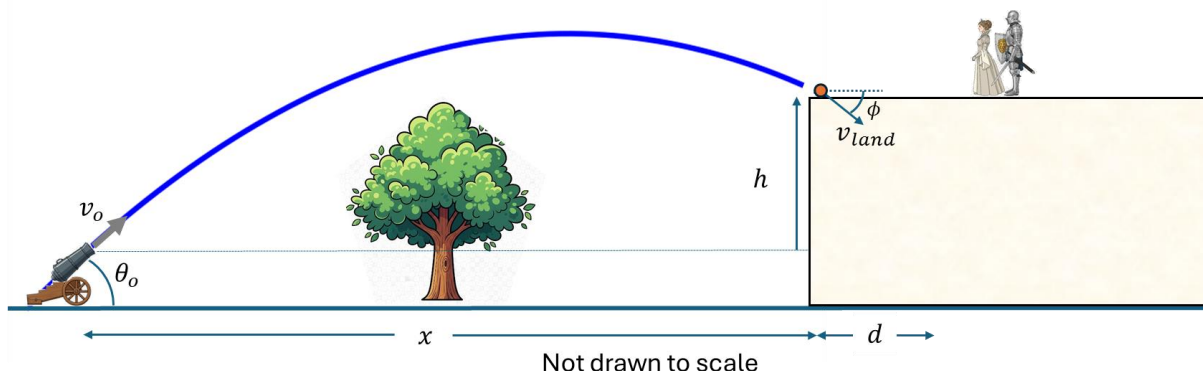
ANS:

g / s

- ii. Explain why liquid cooling is preferred to air cooling.

Q2. [10 points]

As shown in the figure below, a cannon ball was fired at a princess and a soldier on the rooftop balcony of a castle from a hidden position at horizontal distance x . From the damage, it is deduced that the cannon ball landed at a speed of $v_{land} = 34.8 \text{ m/s}$ at an angle $\phi = 33.5^\circ$ below horizontal. From the type of cannon ball, it can be deduced that it is fired with a velocity of $v_o = 40 \text{ m/s}$. You are further told that the rooftop is $h = 20 \text{ m}$ above the level of the cannon and that the cannon ball landed 4.77 s after the sound of the cannon firing was heard (you can ignore the time taken for sound to travel). Ignore air resistance.



- a) Using the information provided,
- Determine x . Leave your answer to 1 d.p.
 - Determine the angle of launch θ_o . Leave your answer 1 d.p.

ANS: $x =$

 m

ANS: $\theta_o =$

- b) The first cannon ball missed and so the attacker will adjust θ_o to have a second try since v_o is fixed. A spy reported back to the princess that the enemy has derived the formula to obtain the optimal angle that will give the horizontal maximum range even if the launch height and landing height are different. That formula for optimal angle θ_{opt} is

$$\tan \theta_{opt} = \frac{v_o}{\sqrt{v_o^2 - 2gh}}$$

where the v_o and h are as defined in the story above and g is acceleration of free fall.

- Determine the optimal angle θ_{opt} . Leave your answer to 1 d.p.

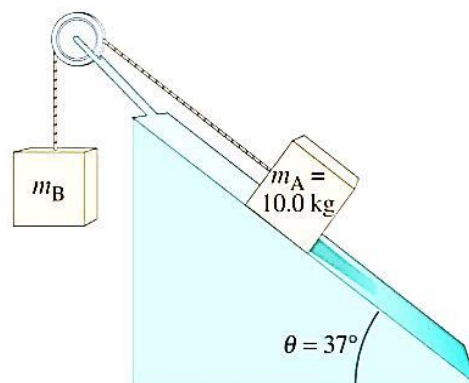
ANS:

- ii. Assuming that the formula is correct, determine the distance d where it would be safe for the princess, not to be hit by the cannon ball. Leave your answer to the nearest m.

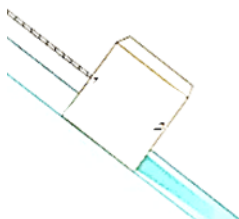
ANS:

Q3. [11 points]

- a) In the figure shown below, two blocks are connected by a massless, inextensible string over a frictionless pulley. The mass of block A is $m_A = 10.0 \text{ kg}$ and the mass m_B of block B is unknown. The block with mass m_A is on the incline with $\theta = 37^\circ$. The coefficient of kinetic friction between the block with mass m_A and the incline is $\mu_k = 0.25$. The system of blocks has an acceleration of 0.80 m/s^2 with m_B moving downwards.



- i. Using the diagram below, indicate the forces acting on block A (with labelled arrows).

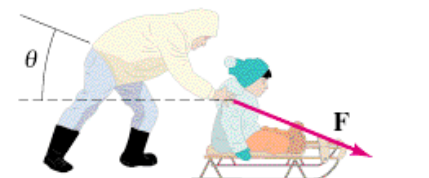


- ii. Calculate mass m_B of block B. Leave your answer to 1 d.p.

ANS:

kg

- b) A man is pushing a boy on a sledge with a force of $F = 200 \text{ N}$ at an angle $\theta = 30^\circ$ as shown in the diagram. The combined weight of the boy and the sledge is 300 N . The coefficient of kinetic friction between the sledge and the ground is 0.2 . Calculate the acceleration of the boy and the sledge.



ANS:

m / s^2

----- The End -----